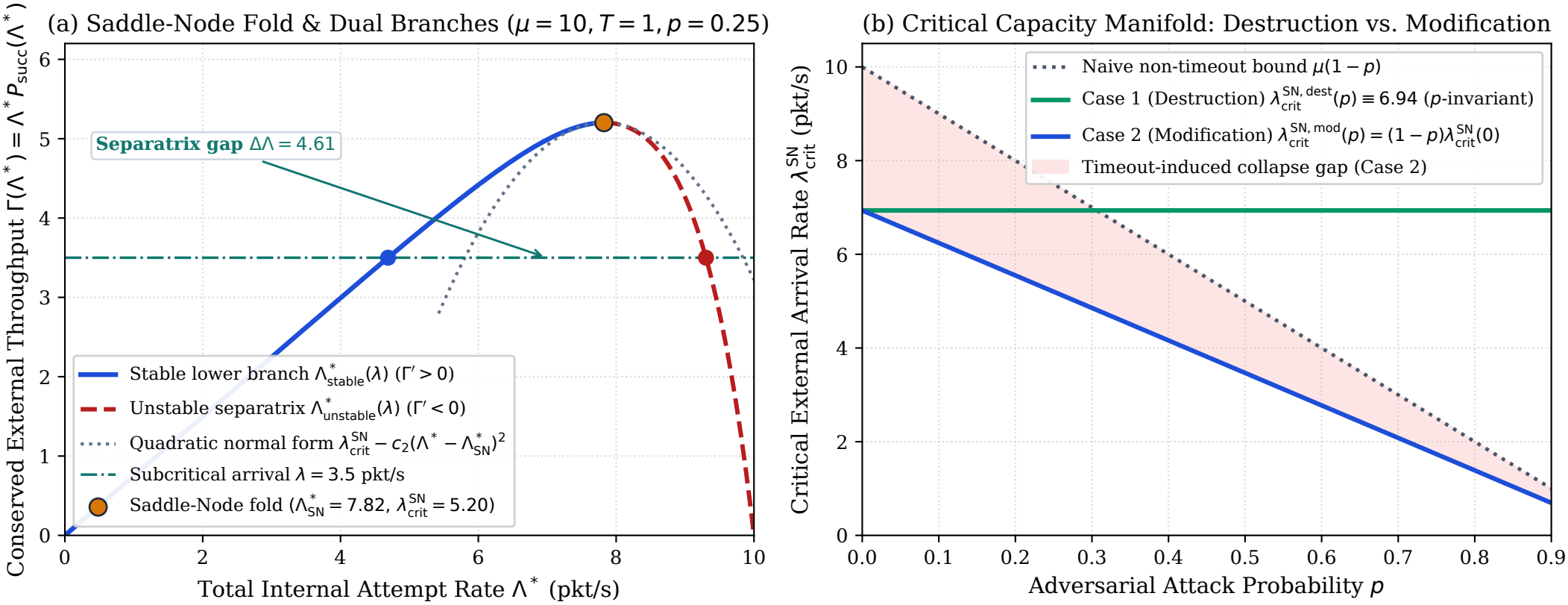
(a) Saddle-Node Fold & Dual Branches ($\mu = 10, T = 1, p = 0.25$)
Conserved External Throughput $\Gamma(\Lambda^*) = \Lambda^* P_{\text{succ}}(\Lambda^*)$
Total Internal Attempt Rate Λ^* (pkt/s)
Separatrix gap $\Delta\Lambda = 4.61$
Stable lower branch $\Lambda^*_{\text{stable}}(\lambda)$ ($\Gamma' > 0$)
Unstable separatrix $\Lambda^*_{\text{unstable}}(\lambda)$ ($\Gamma' < 0$)
Quadratic normal form $\lambda^{\text{SN}}_{\text{crit}} - c_2(\Lambda^* - \Lambda^*_{\text{SN}})^2$
Subcritical arrival $\lambda = 3.5$ pkt/s
Saddle-Node fold ($\Lambda^*_{\text{SN}} = 7.82$, $\lambda^{\text{SN}}_{\text{crit}} = 5.20$)
(b) Critical Capacity Manifold: Destruction vs. Modification
Critical External Arrival Rate $\lambda^{\text{SN}}_{\text{crit}}$ (pkt/s)
Adversarial Attack Probability p
Naive non-timeout bound $\mu(1-p)$
Case 1 (Destruction) $\lambda^{\text{SN, dest}}_{\text{crit}}(p) \equiv 6.94$ (p-invariant)
Case 2 (Modification) $\lambda^{\text{SN, mod}}_{\text{crit}}(p) = (1-p)\lambda^{\text{SN}}_{\text{crit}}(0)$
Timeout-induced collapse gap (Case 2)